

Ying Lin

[linyopt.github.io](https://github.com/linyopt)

Department of Data and Systems Engineering
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RESEARCH INTEREST

► **Machine learning and data science**

Theories, algorithms and applications, especially *sparse learning*, *support vector machine*, *kernel-based learning* and *neural networks*, etc.

► **Continuous Optimization**

Convex and nonconvex optimization: theories, algorithms and applications. Special topics about *error bounds*.

WORK EXPERIENCE

- 2024.11 – present Postdoc Fellow, Department of Data and Systems Engineering, The University of Hong Kong
 Supervisor: Prof. Man-Chung Yue
- 2021.09 – 2024.09 Teaching Assistant, Department of Applied Mathematics, The Hong Kong Polytechnic University

EDUCATION

- 2021.09 - 2024.09 Ph.D., Applied Mathematics, The Hong Kong Polytechnic University, Hong Kong, China.
 Supervisor: Prof. Ting Kei Pong
- 2018.09 - 2021.06 M.Phil., Computational Mathematics, South China Normal University, Guangzhou, China.
 Supervisor: Prof. Dr. Qi Ye
- 2014.09 - 2018.06 BA, Information and Computing Science, South China Normal University, Guangzhou, China.
 Advisor: Prof. Dr. Qi Ye

PREPRINTS

1. **Ying Lin**, Benjamin Poignard. *Change-point detection in variance-covariance matrix*. Submitted May 2026. Code available in my [GitHub repository](#).
2. **Ying Lin**, Tianxiang Liu, Bruno F. Lourenço. *Facial reduction for nice (and non-nice) convex programs*. Submitted December 2025.
3. **Ying Lin**, Yao Kuang, Ahmet Alacaoglu, Michael P. Friedlander. *Decentralized Optimization with Topology-Independent Communication*. Submitted September 2025. JULIA Code available in Michael's [GitHub organization](#).

PUBLICATIONS

1. **Ying Lin**, Benjamin Poignard, Ting Kei Pong, Akiko Takeda. *Change point detection in precision matrices with D -trace loss*. Journal of Computational and Graphical Statistics, 2026, 1–36. MATLAB Code available in my [GitHub repository](#).
2. **Ying Lin**, Scott B. Lindstrom, Bruno F. Lourenço, Ting Kei Pong. *Tight error bounds for log-determinant cones without constraint qualifications*. Journal of Optimization Theory and Applications, 2025, volume 205, article 45.
3. **Ying Lin**, Scott B. Lindstrom, Bruno F. Lourenço, Ting Kei Pong. *Generalized power cones: optimal error bounds and automorphisms*. SIAM Journal on Optimization, 2024, 34(2):1316-1340.
4. **Ying Lin**, Yimin Wei, Qi Ye*. *A Homotopy method for multikernel-based approximation*. Journal of Nonlinear and Variational Analysis, 2022, 6(2):139-154.
5. **Ying Lin**, Qi Ye*. *Support vector machine classifiers by non-Euclidean margins*. Mathematical Foundations of Computing, 2020, 3(4):2-5.
6. **Ying Lin**, Rongrong Lin*, Qi Ye. *Sparse regularized learning in the reproducing kernel Banach spaces with the ℓ^1 norm*. Mathematical Foundations of Computing, 2020, 3(3):205-218.
7. Qi Ye*, **Ying Lin**. *Application of machine learning methods based on LAZE priors in cancer data*. Journal of South China Normal University (Natural Science Edition), 2018, 50(04):115-120.
In Chinese, part of Bachelor's thesis.

TALKS

- 2025.12 *Convex facial reduction algorithms and strong extended duals*
Cones: Theory and Optimization, Kyoto University, Kyoto, Japan
- 2025.11 *Convex facial reduction algorithms and strong extended duals*
APORS Youth Forum, The Hong Kong Polytechnic University, Hong Kong, China
- 2023.06 *Error bounds for the generalized power cone and applications in algebraic structure*
SIAM Conference of Optimization (OP23), Seattle, Washington, U.S.
- 2022.12 *Error bounds for the generalized power cone and applications in algebraic structure*
CAS AMSS-PolyU SIAM Student Chapter Workshop, online
- 2020.11 *Non-Euclidean support vector classifiers for sparse learning*
CSIAM Students Forum 2020, online
- 2020.10 *Non-Euclidean support vector classifiers for sparse learning*
CSIAM 2020, Changsha, Hunan Province, China
- 2019.10 *Support vector machine classifiers by maximum margin of arbitrary norm*
2019 Optimization Frontier Progress Seminar, China West Normal University, Nanchong, Sichuan Province, China

- 2018.10 *The sparse regression model based on LAZE prior – The application in prostate cancer detection*
Information Science Young Scientist Forum, Jinan University, Guangzhou, Guangdong Province, China
- 2018.07 *The sparse regression model based on LAZE prior – The application in prostate cancer detection*
ICSA 2018, Qingdao, Shandong Province, China

VISITING EXPERIENCES

- 2024.01 – 2024.05 Department of Applied Mathematics, The University of British Columbia
Host: [Professor Michael P. Friedlander](#)
- 2023.03 The Institute of Statistical Mathematics
Host: [Professor Bruno F. Lourenço](#)